

Learning 3D Scene Reconstruction from Nighttime Driving Videos

Andrea Ramazzina, Stefano Gasperini, Mario Bijelic, Felix Heide and Federico Tombari

Abstract—Neural Radiance Fields and Gaussian Splatting have emerged as powerful representations for reconstructing complex dynamic urban scenes from multi-view driving logs. By producing photorealistic and geometrically consistent renderings, these methods offer a foundation for closed-loop simulation as well as scalable data augmentation engine, enabling the synthesis of diverse viewpoints and conditions without the need for costly additional data collection in the real world. However, existing approaches are almost exclusively tailored to well-lit, daytime environments and struggle with the challenges of nighttime settings, where noise, strong flares, and moving light sources dominate the visual signal. To address these limitations, we propose NightSplat, a Gaussian Splatting framework specifically designed for high-fidelity reconstruction and novel view synthesis of nighttime driving scenes. Through dedicated components, our method explicitly models sensor noise, lens flare effects, and introduces a lightweight module to represent dynamic vehicle lights. Extensive evaluations on nuScenes and Waymo Open Dataset demonstrate that NightSplat significantly outperforms prior state-of-the-art baselines, both quantitatively and qualitatively, thereby extending simulation and testing to nighttime scenes. Project page: night-splat.github.io

Index Terms—Computer Vision for Transportation, RGB-D Perception.

I. INTRODUCTION

Scene understanding under nighttime conditions is a long-standing challenge in computer vision and robotics [1], [2]. For safety-critical domains such as autonomous driving, nighttime remains one of the most failure-prone operating regimes, as perception models trained predominantly on daytime data often generalize poorly to scenes affected with drastically reduced illumination, strong artificial light sources, and sensor-specific noise [3], [4]. This domain gap can have severe implications, as nighttime driving accounts for a high fraction of traffic accidents [5], [6]. Beyond autonomous vehicles, robust nighttime perception is also crucial for robotics applications such as search-and-rescue and outdoor navigation, where reliable scene understanding is required under unconstrained conditions.

Nighttime environments are particularly challenging because of multiple degradation factors. Reduced illumination lowers the signal-to-noise ratio of captured images, artificial

lights introduce severe brightness contrasts, and lens-related effects such as glare, bloom, and flares distort appearance in a non-linear fashion. These factors not only degrade visual quality but also undermine the geometric consistency needed for multi-view reconstruction. Standard pipelines designed for daylight scenes [7]–[9], therefore, tend to break down when applied to nighttime captures.

With the increasing deployment of autonomous systems in the wild, scalable and domain-gap-free simulation environments for closed-loop evaluation and data generation have become essential. Traditional asset-based simulation pipelines, which rely on artist-designed 3D assets, fall short of the required scale, diversity, and realism. On the other hand, data-driven approaches can reconstruct environments directly from onboard sensor logs [7]–[9], thus offering both realism and scalability.

In this context, Neural Radiance Fields (NeRFs) [10] and Gaussian Splatting (GS) [11] have emerged as state-of-the-art approaches for scene reconstruction and novel view synthesis, relying only on a set of posed captures. These methods have been successfully extended to outdoor driving scenarios by handling moving actors [12]–[16], integrating multiple sensor modalities [17]–[19], or extending robustness to weather conditions [20], [21]. However, most of these advances remain constrained to well-lit daytime scenes. At night, their assumptions break down: the combination of low signal-to-noise ratios, dynamic illumination, and view-dependent artifacts from headlights and streetlights introduces strong inconsistencies across views and prevents faithful reconstructions.

A different line of research has tackled nighttime degradations at the single-image level, including low-light enhancement [22]–[26] and flare removal [27], [28]. While effective for improving visual quality, such methods operate purely in the 2D image space and lack the 3D consistency required for multi-view scene reconstruction and novel view generation. More recent nighttime reconstruction methods have taken steps in this direction [21], [29], [30], but they are limited to static environments and cannot capture the complex light interactions arising from multiple vehicles and streetlights, which is a defining feature of nighttime driving.

To address this gap, we propose NightSplat, a novel Gaussian Splatting framework that extends current approaches for daylight scenes to faithfully handle nighttime sequences. Specifically, we overcome three limiting assumptions in state-of-the-art automotive NeRF/GS pipelines: (1) sensor captures are noise-free, (2) lens flares and other illumination artifacts can be ignored, and (3) lighting is static or globally consistent across frames. Our framework tackles them by modeling acquisition noise in raw sensor space, disentangling flare effects through a lightweight learned module, and efficiently representing dynamic light sources from moving vehicles. Our

Manuscript received: January, 26, 2026; Revised April 13, 2026; Accepted May, 17, 2026.

This paper was recommended for publication by Editor Abhinav Valada upon evaluation of the Associate Editor and Reviewers' comments.

A. Ramazzina is with the Faculty of Computer Science, Technical University of Munich (TUM), Munich, Germany, and with Mercedes-Benz, Stuttgart, Germany (corresponding author andrea.ramazzina@mercedes-benz.com).

S. Gasperini and F. Tombari are with the Faculty of Computer Science, Technical University of Munich (TUM) and Munich Center for Machine Learning (MCML), Munich, Germany.

M. Bijelic and F. Heide are with the Faculty of Computer Science, Princeton University, Princeton, USA, and Torc Robotics, New York, USA.

Digital Object Identifier (DOI): see top of this page.

contributions are summarized as follows:

- We model the scene in the unprocessed image sensor space and jointly learn a per-pixel noise component and camera processing pipeline, ensuring faithful reproduction of acquisition noise while enabling supervision in standard (8bit) sRGB space.
- We model lens flare effects through a lightweight network, decoupling their appearance from the underlying geometry of the scene.
- We introduce a novel neural rendering framework to efficiently represent the contributions of moving light sources in the scene.
- We validate our method on the nuScenes and Waymo datasets, demonstrating state-of-the-art performance.

II. RELATED WORK

A. Radiance Fields for Automotive Data

Early NeRF approaches [31], [32] demonstrated high-fidelity novel view synthesis in outdoor environments by extending the original NeRF formulation to handle unbounded scenes. While these methods achieved high accuracy for static outdoor scenes, they were not designed to handle the temporal and spatial complexity of dynamic driving environments.

To solve this issue, NSG [12] was among the first to extend radiance field representations to automotive datasets by explicitly modeling moving vehicles as separate dynamic entities within a scene graph, whose edges encode the relative positional transformations over time. Subsequent works [9], [13], [33] have improved the reconstruction by enhancing both geometric accuracy and rendering fidelity. In particular, NeuRAD [13] achieved accurate reconstruction by modeling sensor-specific characteristics such as the rolling shutter effect for RGB cameras and beam divergence for LiDAR, paving the way for sensor-realistic closed-loop simulation.

Departing from NeRF-based representation, several works have recently adapted 3DGS [11] for the automotive domain. Notably, Street Gaussians [15] represents the scene as a static background and the moving vehicles as sets of Gaussians with optimizable tracked vehicle poses, while representing the time-varying appearance through spherical harmonics. Subsequent works further extended this approach. Namely, OmniRe [8] integrated an SMPL-based representation to model the pedestrian in the scene, enabling their re-rendering in novel poses. In contrast, SplatAD [14] achieved state-of-the-art results by incorporating accurate sensor simulation of both camera and LiDAR, and modeling view-dependent effects through a lightweight CNN head.

Another line of work focused instead on learning a more robust driving scene representation from captures taken during different times of the day [7], [34], adverse weather [20], [21], or different sensor modalities [17]–[19]. However, while these developments collectively establish a solid foundation for high-fidelity automotive scene reconstruction, they remain focused on well-lit, daytime environments or require captures taken from different lighting conditions. Moreover, none of these methods model the capture noise inherent to low-light imaging, a critical factor in nighttime conditions.

B. Nighttime Image Restoration

A straightforward solution to nighttime reconstruction could be to transform the nighttime captures into daytime images, and then feed them to a daytime reconstruction method. Different enhancement and restoration techniques have been proposed to solve the unique challenges typical of nighttime imaging (low signal-to-noise, color distortion, limited dynamic range).

Low-light enhancement methods [22]–[26] have been developed to brighten and restore dark images. These methods typically employ deep neural networks trained to map low-light RGB images to well-lit, noise-free counterparts, and can be trained either in a supervised manner using paired data [24], [26] or in an unsupervised manner [22], [23].

Other works [27], [28] have addressed different problems arising in nighttime scenarios, such as lens flares caused by strong light sources (e.g., headlights, streetlights). To address this, recent datasets and restoration pipelines focus on flare removal by explicitly modeling flare patterns or disentangling them from the underlying clean image. Such methods either rely on synthetic data augmentation to train robust flare removal networks [27], or integrate flare-aware priors to improve restoration performance in real-world driving conditions [28].

C. Nighttime Scene Reconstruction

In parallel, another line of work focused on recovering accurate 3D scene geometry from night images [1]–[4]. These works [2]–[4] are usually centered on generating nighttime data with available depth information and use this data to train a deep learning network to infer the scene depth, or treat it as a domain adaptation problem and focus on the extraction of daylight-invariant features [1].

Inspired by the success of radiance fields in daylight scene reconstructions, recent works have also proposed to extend NeRF and GS methods for more accurate 3D reconstruction. Notably, RawNeRF [30] and HDRsplat [35] operate directly on the raw (unprocessed) RGB captures, allowing for the disentanglement of acquisition and quantization noise. However, such raw captures are not readily available in automotive settings, due to the prohibitive storage requirements and the common practice of collecting only ISP-processed 8-bit images. Other works instead focused on directly reconstructing the scene from 8-bit images. Among these, Gaussian-DK [29] addresses the multi-view inconsistencies by modeling the camera response and learning a set of light features, while ResorGS [21] leverages a formation model based on a physically inspired attenuation map to recover the underlying scene. While these methods demonstrate promising advances in nighttime scene reconstruction, they rely on controlled capture conditions [30], [35] or simplified lighting interactions [21], [29] and are hence unfit for large-scale automotive settings, characterized by complex, dynamic illumination effects (e.g., moving headlights, reflections, and glare). In contrast, our approach works on in-the-wild outdoor scenes and is able to handle complex lighting effects common in automotive scenes.

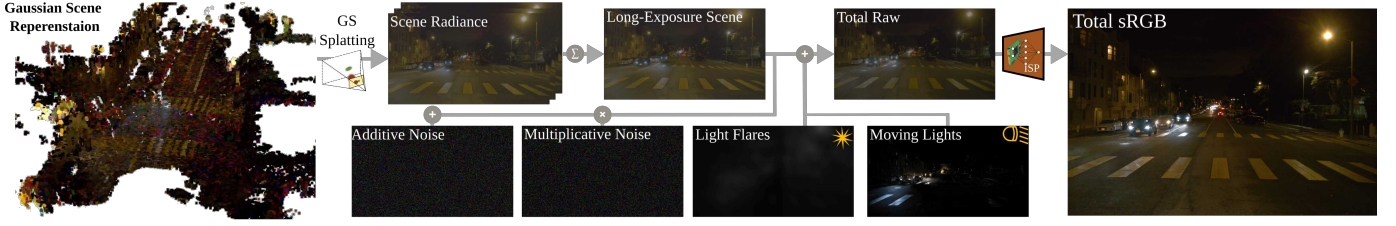


Fig. 1: **Overview.** We represent the scene as a set of 3D Gaussians that we splat to render the scene radiance in raw sensor space and model the long-exposure settings typical of nighttime captures by fusing multiple splats (Sec. III-A). Then we include acquisition noise and the effects of light sources (Sec. III-B) to obtain the final raw capture. This is finally transformed with a differentiable ISP pipeline (Sec. III-C) to the image that we use to supervise the reconstruction (Sec. III-D).

III. NIGHTSPLAT

We represent the scene with a set of 3D Gaussians, encoding the base color appearance (in linear raw RGB space) and surface properties. We compute the scene rendering I_P at camera position P by first splatting the Gaussians on the target plane, described in Sec. III-A, and explicitly modeling the influence of the moving light sources and flares via two lightweight CNNs f_l and f_{flare} , as outlined in Sec. III-B. We finally convert the rendered capture to an 8-bit noisy image as described in Sec. III-C and use this as the supervision signal during training, detailed in Sec. III-D. An overview illustration of our method is provided in Figures 1 and 2.

A. Scene Representation

We represent the scene as a set of 3D Gaussians [11], each characterized by a learnable opacity $o \in (0, 1)$, mean $\mu \in \mathbb{R}^3$, and covariance matrix $\Sigma \in \mathbb{R}^{3 \times 3}$, parametrized by a quaternion $\mathbf{q} \in \mathbb{R}^4$ and scaling factors $\mathbf{s} \in \mathbb{R}_+^3$. Each Gaussian also has a learned color appearance $\mathbf{c} \in \mathbb{R}^{N_{sh}}$ —represented as N_{sh} spherical harmonics coefficients as in [11]—and a learnable reflectance feature $\zeta \in \mathbb{R}^{N_\zeta}$ to model how the surface reflects the light.

1) *Scene Graph Modeling:* To model dynamic traffic participants, following previous works [8], we instantiate a scene graph composed of three classes: static scene, vehicles, and deformable objects. The position and rotation of each Gaussian assigned to a vehicle instance v is defined locally with respect to the vehicle’s bounding box center. To render an image at timestep t , each of these Gaussians is then rotated and translated using the (known) vehicle position $\mathbf{V}_{v,t} \in \mathbb{SE}(3)$. Each vehicle v has also associated two translation vectors $\mathbf{t}_{v,1}, \mathbf{t}_{v,2} \in \mathbb{R}^3$ and embeddings $\mathbf{l}_{v,1}, \mathbf{l}_{v,2} \in \mathbb{R}^L$, which we add to respectively describe the relative position of the front headlights and their appearance. Analogously, the ego vehicle is also associated with two translation vectors $\mathbf{t}_{c,1}, \mathbf{t}_{c,2}$ (describing the relative light position from the front-camera) and embeddings $\mathbf{l}_{c,1}, \mathbf{l}_{c,2} \in \mathbb{R}^L$.

For deformable objects, such as pedestrians or cyclists, we rely on a deformation network f_ϕ as in [8], predicting per-Gaussian translations and rotations across time. This allows us to capture dynamic motions that cannot be represented by simple rigid transformations. Unlike SMPL-based human models [36], which often fail under low-light conditions due

to unreliable pose estimation, this approach provides a more robust and general representation for nighttime conditions.

2) *Scene Splatting:* To render the radiance $C_{P,t}(x, y)$ observed at pixel (x, y) for camera pose P at timestep t , we follow the standard α -blending procedure of 3DGS [11]. The Gaussians contributing to each pixel are first sorted by their depth relative to the camera center and then composited:

$$\hat{C}_{P,t} = \sum_{i \in \mathcal{N}} c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j) + (1 - \sum_{i \in \mathcal{N}} \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j)) C_{bck} \quad (1)$$

where $\alpha_i = o_i \exp(-\frac{1}{2}(P - \mu_i)^\top \Sigma_i^{-1} (P - \mu_i))$ is the opacity and $C_{bck} \in [0, 1]^{H \times W \times 3}$ denotes a learned background map that accounts for sky and far-range regions.

In addition to radiance, we splat per-Gaussian features ζ to obtain a feature map F_P , which is subsequently used to model moving light sources. We also render the depth map D_P as well as the normal map N_P , computed by splatting the shortest axis of each Gaussian, flipped if not facing the camera, following [37].

Note that, to simplify the notation, the temporal variable t is omitted when not explicitly needed.

3) *Exposure Modeling:* To account for low light conditions at nighttime, the camera integrates incoming photons over a longer period of time. However, as the ego and the other vehicles moves during this process, it produces visible motion blur in real world captures. To account for this effect, unlike [14], we explicitly approximate the exposure as a temporal integration between adjacent frames. Specifically, given the camera poses at the previous and next timesteps, referred as P^{-1} and P^{+1} , we render the radiance captures at intermediate camera positions:

$$\hat{C}_{P^-} = \hat{C}_{P+\delta(P^{-1}-P)}, \quad \hat{C}_{P^+} = \hat{C}_{P+\delta(P^{+1}-P)},$$

where $\delta \in [0, 1]$ is a learnable interpolation factor that controls the effective exposure span. The final exposure-aware radiance is then obtained by per-pixel averaging, i.e.:

$$C_P = \frac{1}{3} (\hat{C}_P + \hat{C}_{P^-} + \hat{C}_{P^+}).$$

B. Light Modeling

1) *Flare Effects:* In addition to motion blur, long exposure at night also accentuates flare artifacts around bright light sources (e.g., street lights), caused by lens scattering and

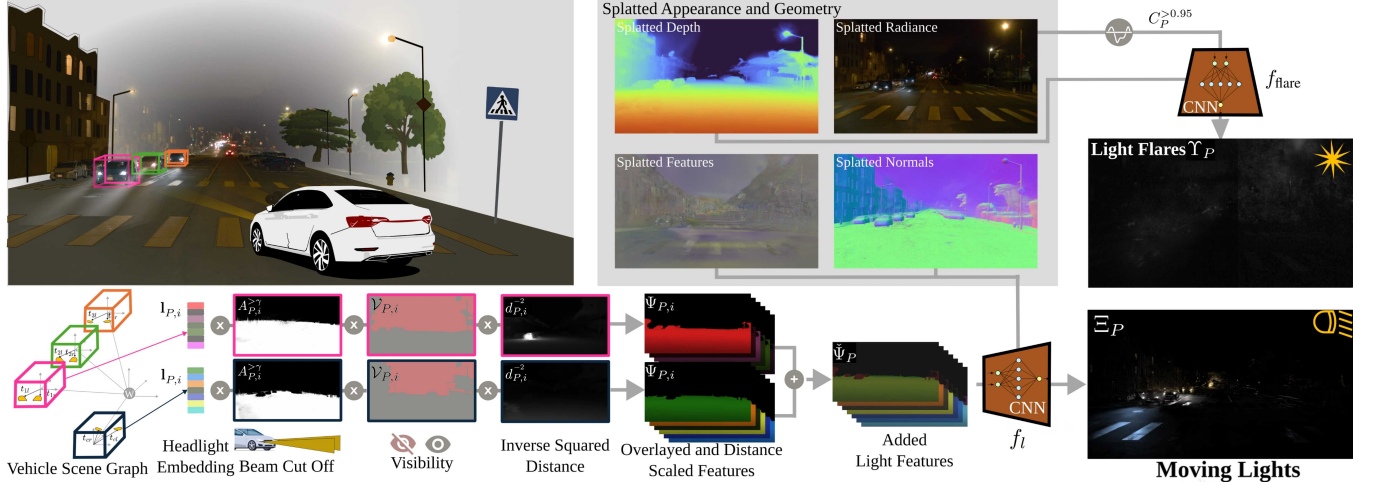


Fig. 2: **Light modeling.** Each dynamic light source i is represented by an appearance embedding $\mathbf{l}_i$, and its relative position $\mathbf{t}_i$. The light feature $\Psi_{P,i}$ is obtained by masking the tiled appearance embedding $\mathbf{l}_i^{\text{tile}}$ with the beam cut-off angle $A_{P,i}^{>\gamma}$ and the visible scene content $\mathcal{V}_{P,i}$, and scaled by the inverse square surface distance $d_{P,i}^{-2}$. The summed light features $\check{\Psi}_P$ are processed by a neural network f_l , along with the splatted scene features F_P and surface normals N_P , to predict the illumination effects Ξ_P . To model the lens flares, we use a lightweight CNN network f_{flare} that estimates the flares present in the capture using the splatted depth and scene radiance, thresholded to keep only the saturated areas.

sensor blooming [27]. These effects cannot be captured by Gaussian splatting alone, as they depend on both light intensity and optical distortions. We model this phenomenon via a shallow CNN f_{flare} that operates only on the regions most likely to correspond to bright lights. Concretely, we extract a high-intensity image $C_P^{>0.95}$ from the splatted image C_P , where pixels with values ≤ 0.95 are set to zero and all other pixels retain their original values. We integrate the geometry information via the splatted depth map, feeding $C_P^{>0.95}$ and D_P in f_{flare} to produce the flare layer:

$$\Upsilon_P = f_{\text{flare}}([C_P^{>0.95} \mid D_P]),$$

where $[\cdot \mid \cdot]$ denotes channel-wise concatenation.

2) *Moving Light Sources:* We separately model the effects of light sources on moving vehicles (shown in Fig. 3) because of their role in the appearance of the environment. We represent these lights with respect to each vehicle in the scene graph introduced in Sec. III-A. Their contribution depends on their relative position to the scene and the camera, modeled through a CNN network f_l , as shown in Fig. 2.

Specifically, for each light source i , we associate a learnable embedding vector $\mathbf{l}_i \in \mathbb{R}^D$ that encodes its appearance. Its contribution to a pixel is modulated by three geometric factors: (i) a visibility term $\mathcal{V}_{P,i}(x, y) \in [0, 1]$, (ii) an inverse-square distance falloff $d_{P,i}(x, y)^{-2}$, and (iii) a cutoff term $A_{P,i}^{>\gamma}(x, y)$ that accounts for the limited vertical spread of headlights, modeled as a function of the elevation angle of the surface point w.r.t. the light. The per-light contribution at pixel (x, y) is then defined as the Hadamard product:

$$\Psi_{P,i}(x, y) = \mathbf{l}_i^{\text{tile}}(x, y) \mathcal{V}_{P,i}(x, y) d_{P,i}(x, y)^{-2} A_{P,i}^{>\gamma}(x, y). \quad (2)$$

We obtain $d_{P,i}$ by computing the Euclidean distance between the light source position $\mathbf{o}_i = V_i^T \mathbf{t}_i$ and each Gaussian center $\boldsymbol{\mu}$ and then get the per-pixel depths during the scene splatting as in Eq. 1. The cutoff term $A_{P,i}^{>\gamma}(x, y)$ is similarly obtained by

computing the per-Gaussian angle $a = \text{atan2}(\boldsymbol{\mu}_y - \mathbf{o}_y, \boldsymbol{\mu}_x - \mathbf{o}_x)$, then splatting it on image space and thresholding it at a fixed value γ . The visibility term $\mathcal{V}_{P,i}$ is obtained analogously to [38].

The contributions from all light sources are then summed:

$$\check{\Psi}_P(x, y) = \sum_i \Psi_{P,i}(x, y).$$

We then concatenate $\check{\Psi}_P$ with the splatted reflectance features F and splatted normals N (for additional geometric context), and process the result with a lightweight CNN f_l :

$$\Xi_P(x, y) = f_l([\check{\Psi}_P(x, y) \mid F_P(x, y) \mid N_P(x, y)]),$$

where $[\cdot \mid \cdot]$ denotes channel-wise concatenation.

Finally, total radiance capture is then computed as:

$$I_P(x, y) = C_P(x, y) + \Upsilon_P(x, y) + \Xi_P(x, y).$$

C. rawRGB-to-sRGB Modeling

To bridge the gap between physically-motivated rendering and standard sRGB supervision, we model a fully differentiable camera processing pipeline that converts I , generated in linear raw sensor space, into an sRGB image I^{sRGB} . This pipeline is designed to emulate the essential processing stages of a modern digital camera, incorporate realistic sensor noise characteristics, and remain fully differentiable for gradient backpropagation.

1) *Noise Modeling:* For each camera sensor j , to better align with real camera image statistics, we include two terms modeling the noise commonly present during the acquisition process [39], namely the shot noise $\mathcal{P}$, modeled with a Poisson distribution, and the fixed pattern noise, here represented by the per-pixel multiplicative and additive terms $M_j(x, y)$, $B_j(x, y)$, obtaining the noised image I^n as:

$$I_{P,j}^n(x, y) = \mathcal{P}(M_j(x, y)I_P(x, y) + B_j(x, y)). \quad (3)$$

We initialize the matrices M_j and B_j respectively with unity and zero values, and optimize them during training. We apply the shot noise $\mathcal{P}$ only at inference time, as its zero-mean fluctuations average out across views during training.

2) *In-camera processing*: After adding noise, we employ a lightweight, fully differentiable processing pipeline to obtain the sRGB image $I^{\text{sRGB}}_{P,j}$, which we use for supervision with the ground-truth captures. This pipeline comprises three steps: white balancing, color space transformation, and gamma encoding. It is formulated as:

$$\begin{aligned} I^{\text{wb}}_{P,j}(x, y) &= I^n_{P,j}(x, y) \odot \mathbf{g}_{t,j}, \\ I^{\text{cc}}_{P,j}(x, y) &= H_{t,j} I^{\text{wb}}_{P,j}(x, y), \\ I^{\text{sRGB}}_{P,j}(x, y) &= \text{clip}(\mathcal{G}(I^{\text{cc}}_{P,j}(x, y)), 0, 1), \end{aligned} \quad (4)$$

where $\odot$ denotes element-wise multiplication across color channels with the white balance gains $\mathbf{g}_{t,j} \in \mathbb{R}^3$, $H_{t,j} \in \mathbb{R}^{3 \times 3}$ is the color correction matrix, the superscripts *wb* and *cc* denote white-balanced and color-corrected terms, and $\mathcal{G}(\cdot)$ denotes the standard sRGB gamma transfer function [40]. We initialize all parameters such that the raw-to-sRGB mapping is initially the identity, and optimize them jointly with the scene representation during training.

D. Training Supervision

We supervise our representation via a reconstruction loss $\mathcal{L}_c$ with the available ground-truth sRGB captures. Due to the limited information present in the photometric loss, we also constrain the learned geometry with $\mathcal{L}_{\text{geom}}$ and $\mathcal{L}_{\text{reg}}$. We regularize the learned camera-processing pipeline with the loss terms $\mathcal{L}_{c_1}, \mathcal{L}_{c_2}$. For clarity, we omit the sensor index j when it is not explicitly required.

1) *Photometric Loss*: We supervise with the ground truth capture frames $\bar{I}$ using a combination of pixel-wise and perceptual similarity (SSIM) [41] terms:

$$\mathcal{L}_c = \sum_P \lambda_1 \|I_P^{\text{sRGB}} - \bar{I}\|_F^2 + \lambda_2 (1 - \text{SSIM}(I_P^{\text{sRGB}}, \bar{I})), \quad (5)$$

2) *Camera Regularization*: We penalize large variations of white balance gains $\mathbf{g}_{t,j}$ and the color correction matrix $H_{t,j}$ between two consecutive timeframes $t, t+1$:

$$\mathcal{L}_{c_1} = \lambda_3 \sum_{t,j} \|\mathbf{g}_{t+1,j} - \mathbf{g}_{t,j}\|_2^2 + \lambda_3 \|H_{t+1,j} - H_{t,j}\|_F^2, \quad (6)$$

where $\|\cdot\|_F$ denotes the Frobenius norm. We also enforce the noise terms M_j, B_j of each camera sensor to be unbiased:

$$\mathcal{L}_{c_2} = \lambda_4 \frac{1}{|\mathcal{N}|} \sum_j (|\sum_{x,y \in \mathcal{N}} M_j(x, y)|^2 + |\sum_{x,y \in \mathcal{N}} B_j(x, y)|^2) \quad (7)$$

where $\mathcal{N}$ denotes a set of local image patches.

3) *Geometry Supervision*: Due to the scarce photometric supervision, we supervise the splatted normals N using pseudo ground truth $\bar{N}$ obtained from an off-the-shelf method [42], and the splatted depth D using the projected LiDAR scan $\bar{D}$ as in [8], [15]:

$$\mathcal{L}_{\text{geom}} = \lambda_5 \sum_P \|D_P - \bar{D}_P\|_2^2 + \sum_P \|N_P - \bar{N}_P\|_2^2, \quad (8)$$

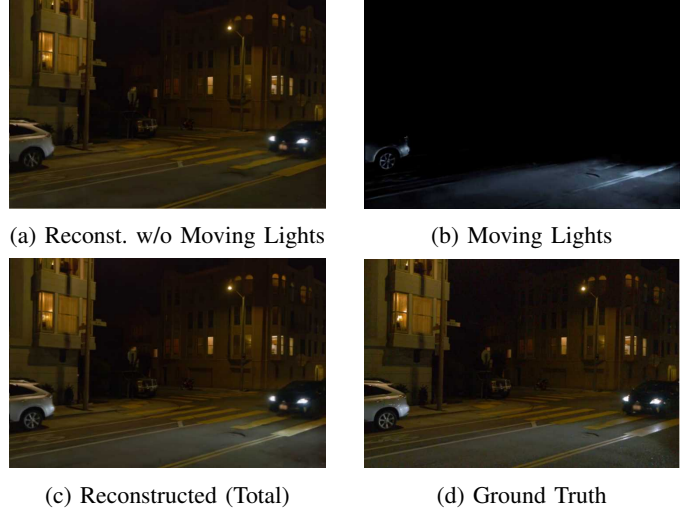


Fig. 3: Visualization of our explicit disentangling of the vehicles' lighting effects (b) from the underlying scene (a).

4) *Geometry Regularization*: We regularize the scene by promoting the 3D Gaussians to progressively become planar [43], and limiting the splatted normals to diverge from the normal map $\hat{N}_P$ derived from the splatted depth:

$$\mathcal{L}_{\text{reg}} = \lambda_6 \sum_u \|\min(\mathbf{s}_u)\| + \lambda_7 \sum_P \|N_P - \hat{N}_P\|_2^2, \quad (9)$$

Finally, the total training loss is obtained as:

$$\mathcal{L}_{\text{tot}} = \mathcal{L}_c + \mathcal{L}_{c_1} + \mathcal{L}_{c_2} + \mathcal{L}_{\text{geom}} + \mathcal{L}_{\text{reg}}, \quad (10)$$

where λ_* are fixed training hyperparameters.

IV. EXPERIMENTS AND RESULTS

A. Implementation Details

We train our NightSplat representation for 20k steps, using the Adam optimizer. As loss parameters, we follow when possible [8] and use $\lambda_1 = 0.8, \lambda_2 = 0.2, \lambda_3 = 10^{-4}, \lambda_4 = 1, \lambda_5 = 10^{-2}, \lambda_6 = 10^{-2}, \lambda_7 = 10^{-3}$. The networks f_{flare} and f_l are composed of two residual blocks with hidden dimensions respectively of 7 and 32, and are optimized with a learning rate of $l_r = 10^{-3}$, while for the light embeddings $\mathbf{l}_*$ it is $l_r = 5 \cdot 10^{-3}$. Unless otherwise specified, we follow the initialization and learning rate scheduling of [8]. We train on a single RTX A6000 GPU for approximately 5 hours, achieving a per-frame rendering speed of 19 Hz at inference.

B. Datasets & Metrics

We follow previous work [8], [15] and evaluate our method on two popular driving datasets containing night scenes, namely NuScenes [44] and Waymo Open Dataset (WOD) [45]. Specifically, for each dataset, we extract eight sequences recorded at nighttime, each consisting on average of 190 timeframes. We utilize all available cameras (6 for NuScenes and 5 for WOD) and 1 LiDAR. For testing, we use a 20% holdout test set, evenly sampled from the sequence. As evaluation metrics, we employ the standard PSNR, SSIM, and LPIPS. Additionally, to quantify the reconstruction quality of the dynamic regions, we compute the PSNR and SSIM over the pixels within the moving-vehicle bounding boxes.

C. Baselines

We compare our approach against seven GS methods, a NeRF approach, and one diffusion-based work, which together represent the state-of-the-art for reconstruction and NVS of automotive scenes. Specifically, we employ four GS-methods and one NeRF tailored to dynamic urban scenes, namely Street Gaussians [15], Periodic Vibration Gaussian (PVG) [16], OmniRe [8], SplatAD [14], NeurAD [13], and two GS methods specific for night-time scenes [21], [29] implemented on top of [15]. Additionally, we also use a method trained for nighttime depth estimation as additional geometry supervision (as in [46]). We use and adapt the code of the GS-based methods from DriveStudio [8]. Finally, to include non-GS approaches for novel view synthesis, we also compare with FreeVS [47], a diffusion model method operating on projected colored point clouds for novel view synthesis. We employ md4all [3] on nuScenes and FreeVS [47] on WOD, as they were trained only on those datasets.

	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Vehicle PSNR $\uparrow$	Vehicle SSIM $\uparrow$
Gaussian-DK [29]	27.82	0.678	0.439	17.04	0.437
RestorGS [21]	27.65	0.675	0.454	16.50	0.449
Street Gaussians [15]	27.67	0.686	0.430	16.45	0.446
OmniRe [8]	27.43	0.678	0.405	19.53	0.533
OmniRe+md4all [3]	25.71	0.641	0.461	18.19	0.437
PVG [16]	28.03	0.690	0.489	17.11	0.476
NeurAD [13]	27.35	0.671	0.495	19.95	0.482
SplatAD [14]	28.79	0.690	0.479	21.12	0.572
NightSplat (Ours)	29.24	0.799	0.197	21.75	0.686

TABLE I: NVS results for nuScenes [44] dataset. Best, second-best, and third-best results are highlighted in red, orange, and yellow, respectively.

	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Vehicle PSNR $\uparrow$	Vehicle SSIM $\uparrow$
Gaussian-DK [29]	28.91	0.704	0.497	21.63	0.586
RestorGS [21]	28.72	0.706	0.503	21.78	0.622
Street Gaussians [15]	28.90	0.713	0.488	21.51	0.600
OmniRe [8]	28.69	0.707	0.468	21.96	0.591
FreeVS [47]	25.12	0.729	0.327	21.91	0.671
PVG [16]	28.91	0.715	0.543	19.08	0.556
NeurAD [13]	28.89	0.714	0.486	22.43	0.601
SplatAD [14]	29.10	0.720	0.475	23.31	0.625
NightSplat (Ours)	29.99	0.736	0.463	24.21	0.690

TABLE II: NVS results for Waymo Open Dataset [45].

D. Results

We report the reconstruction results quantitatively in Tables I, II, and qualitatively in Fig. 5.

1) *Quantitative Results*: Overall, NightSplat consistently outperforms the compared baselines across the reported metrics. On nuScenes (Table I), our method achieves a mean SSIM of 0.799, improving by 0.109 over the next-best methods. Vehicle-centric metrics show even larger relative gains, with the Vehicle SSIM metric improving by 0.114, indicating that our explicit modeling of moving lights and flares yields significant benefits for reconstructing dynamic actors, often the hardest parts of the scene to reconstruct.

This improvement over competing methods is also noticeable on the Waymo Open Dataset (Table II), where our method improves over the next-best method by 0.89 dB in PSNR. FreeVS [47] achieves a lower LPIPS but, as visible in the supplement video, it introduces visual hallucinations

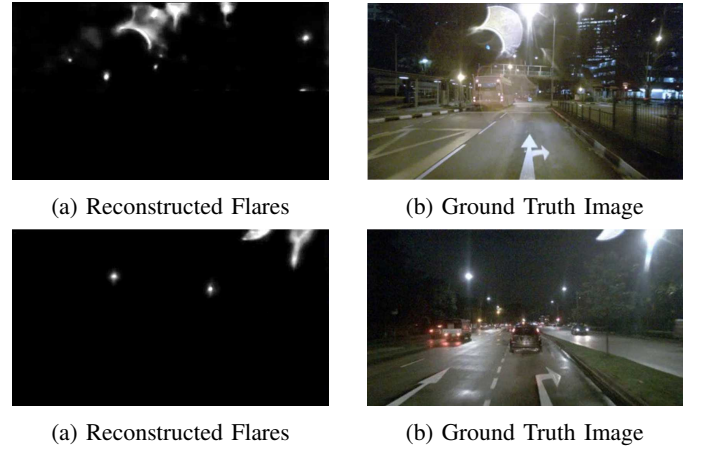


Fig. 4: Visualization examples of our explicit disentangling of the lens flares (a) from the underlying scene (b).

(especially in areas with sparse LiDAR points) and thus yields the lowest PSNR score across all the methods. These consistent gains across datasets demonstrate that the proposed modules generalize across sensors and environments.

We attribute the performance increase gap between WOD and nuScenes to the differences in nighttime imaging conditions and dataset characteristics.

2) *Qualitative Results*: Fig. 5 illustrates qualitative comparisons between our method and competing baselines on both nuScenes (rows 1-3) and the Waymo Open Dataset (rows 4-6). Our approach consistently recovers sharper scene details in both dusk and fully nighttime conditions, where alternative methods often suffer from blur or loss of structure. As highlighted in the zoomed-in regions, our method is particularly effective at capturing characteristic nighttime effects such as lens flares (as in rows 2 and 3), as well as complex interactions of vehicle headlights with the surrounding environment (visible in rows 1,5,6), leading to more accurate and realistic reconstructions.

The effectiveness of our proposed components is further highlighted in Fig. 3 and Fig. 4, where we respectively visualize the disentangled representations of moving light reflections and lens flares. Unlike our approach, baseline methods are unable to explicitly separate and represent these effects, often baking them into the surface appearance or introducing semi-opaque floating artifacts that degrade both visual realism and geometric consistency, as also visible in Fig. 5 for the three shown baselines.

Exposure Modeling	Raw & Noise	Flare Effects	Moving Lights	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
$\times$	$\times$	$\times$	$\times$	26.19	0.652	0.426
$\checkmark$	$\times$	$\times$	$\times$	26.55	0.665	0.443
$\checkmark$	$\checkmark$	$\times$	$\times$	27.22	0.762	0.226
$\checkmark$	$\checkmark$	$\checkmark$	$\times$	27.34	0.765	0.225
$\checkmark$	$\checkmark$	$\checkmark$	$\sim$	27.40	0.763	0.226
$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	27.83	0.776	0.228

TABLE III: Components ablation on a subset of the nuScenes [44] dataset. We use $\sim$ to denote partial inclusion.



Fig. 5: **Qualitative comparison on nuScenes (rows 1–3) and the Waymo Open Dataset (rows 4–6).** Our method (NightSplat) consistently reconstructs sharper nighttime scene details compared to competing methods, reducing blur and scene artifacts, as also highlighted in the zoom-ins (squared in red).

E. Ablation Experiments

To quantify the impact of each proposed component, we run a series of ablation experiments on a subset of the nuScenes nighttime sequences. Table III reports results when progressively enabling the different modules, respectively, exposure-aware blurry modeling, rawRGB representation with noise modeling, flare effects, and moving lights.

We start from a GS-baseline [8], with limited performances as it cannot cope with nighttime degradations. Adding exposure modeling (Sec.III-A3) improves performance moderately (PSNR 26.55dB), highlighting the importance of accounting for long exposures when the ego-vehicle moves. Incorporating raw-space rendering and explicit noise modeling (Sec. III-C) provides a substantial jump in reconstruction quality (PSNR 27.22dB), showing that modeling acquisition characteristics is crucial to reduce inconsistencies with real nighttime captures. Introducing flare modeling on top yields further improvements (PSNR 27.34dB), demonstrating that lens-related artifacts cannot be trivially absorbed into scene appearance and benefit from explicit disentanglement. Naively adding our moving-lights module with $\Psi_P = d_P^{-2}$ (denoted by the $\sim$ symbol in Tab. III) only slightly improves the results. However, with the full formulation (as in Eq. 2), we achieve the best results in

PSNR (27.83 dB) and SSIM (0.776), validating the need to explicitly model the contribution of dynamic light sources.

V. CONCLUSION

In this work we presented NightSplat, a 3D Gaussian Splatting framework specifically designed for high-fidelity reconstruction and novel view synthesis of nighttime driving scenes. Our framework introduces three key novelties: (i) raw-space modeling with sensor noise integration, (ii) a lightweight flare disentanglement module, and (iii) an efficient representation of dynamic light sources from moving vehicles. Extensive experiments on nuScenes and the Waymo Open Dataset demonstrate that NightSplat significantly outperforms existing NeRF- and GS-based baselines, both quantitatively and qualitatively. We believe NightSplat fills an important gap in data-driven digital twin pipelines for autonomous driving: by extending high-quality reconstruction to nighttime conditions, it enables more realistic closed-loop simulation and safer evaluation of perception and planning modules. In future work, we aim to extend our framework to other challenging conditions, such as adverse weather and multi-sensor fusion.

VI. ACKNOWLEDGEMENTS

Andrea Ramazzina has been supported by the German Ministry BMW within the project “NXT GEN AI METHODS”. Stefano Gasperini has been supported by the German Ministry BMFTR under the Robotics Institute Germany (RIG). Felix Heide was supported by an NSF CAREER Award (2047359), a Packard Foundation Fellowship, a Sloan Research Fellowship, a Sony Young Faculty Award, a Project X Innovation Award, a Amazon Science Research Award, and a Bosch Research Award.

REFERENCES

- [1] M. Vankadari, S. Garg, A. Majumder, S. Kumar, and A. Behera, “Unsupervised monocular depth estimation for night-time images using adversarial domain feature adaptation,” in *ECCV*, 2020, pp. 443–459.
- [2] A. Sharma, L.-F. Cheong, L. Heng, and R. T. Tan, “Nighttime stereo depth estimation using joint translation-stereo learning: Light effects and uninformative regions,” in *3DV*. IEEE, 2020, pp. 23–31.
- [3] S. Gasperini, N. Morbitzer, H. Jung, N. Navab, and F. Tombari, “Robust monocular depth estimation under challenging conditions,” in *Proceedings of the IEEE/CVF ICCV*, 2023, pp. 8177–8186.
- [4] F. Tosi, P. Z. Ramirez, and M. Poggi, “Diffusion models for monocular depth estimation: Overcoming challenging conditions,” in *ECCV*. Springer, 2024, pp. 236–257.
- [5] F. H. A. U.S. Department of Transportation, “How do weather events affect roads?” [Online]. Available: <https://ops.fhwa.dot.gov/weather/roadimpact.htm>
- [6] E. R. S. Observatory, “Annual statistical report on road safety in the eu 2021.” [Online]. Available: https://road-safety.transport.ec.europa.eu/system/files/2022-05/ERSO_annual_report_20220509.pdf
- [7] T. Fischer, J. Kulhanek, S. R. Bulò, L. Porzi, M. Pollefeys, and P. Kotschieder, “Dynamic 3d gaussian fields for urban areas,” in *NeurIPS*, 2024.
- [8] Z. Chen, J. Yang, J. Huang, R. de Lutio, J. M. Esturo, B. Ivanovic, O. Litany, Z. Gojcic, S. Fidler, M. Pavone *et al.*, “Omnire: Omni urban scene reconstruction,” *arXiv preprint arXiv:2408.16760*, 2024.
- [9] Z. Wang, T. Shen, J. Gao, S. Huang, J. Munkberg, J. Hasselgren, Z. Gojcic, W. Chen, and S. Fidler, “Neural Fields Meet Explicit Geometric Representations for Inverse Rendering of Urban Scenes,” in *2023 IEEE/CVF CVPR*, pp. 8370–8380.
- [10] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, “Nerf: Representing scenes as neural radiance fields for view synthesis,” *Communications of the ACM*, vol. 65, pp. 99–106, 2021.
- [11] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3d gaussian splatting for real-time radiance field rendering,” *ACM Trans. Graph.*, vol. 42, no. 4, pp. 139–1, 2023.
- [12] J. Ost, F. Mannan, N. Thuerey, J. Knodt, and F. Heide, “Neural scene graphs for dynamic scenes,” in *Proceedings of the IEEE/CVF CVPR*, 2021, pp. 2856–2865.
- [13] A. Tonderski, C. Lindström, G. Hess, W. Ljungbergh, L. Svensson, and C. Petersson, “Neurad: Neural rendering for autonomous driving,” in *Proceedings of the IEEE/CVF CVPR*, 2024, pp. 14 895–14 904.
- [14] G. Hess, C. Lindström, M. Fatemi, C. Petersson, and L. Svensson, “Splatad: Real-time lidar and camera rendering with 3d gaussian splatting for autonomous driving,” in *CVPR*, 2025, pp. 11 982–11 992.
- [15] Y. Yan, H. Lin, C. Zhou, W. Wang, H. Sun, K. Zhan, X. Lang, X. Zhou, and S. Peng, “Street gaussians: Modeling dynamic urban scenes with gaussian splatting,” in *ECCV*. Springer, 2024, pp. 156–173.
- [16] Y. Chen, C. Gu, J. Jiang, X. Zhu, and L. Zhang, “Periodic vibration gaussian: Dynamic urban scene reconstruction and real-time rendering,” *arXiv preprint arXiv:2311.18561*, 2023.
- [17] D. Borts, E. Liang, T. Brodermann, A. Ramazzina, S. Walz, E. Palladin, J. Sun, D. Brueggemann, C. Sakaridis, L. Van Gool *et al.*, “Radar fields: Frequency-space neural scene representations for fmcw radar,” in *ACM SIGGRAPH 2024 Conference Papers*, 2024, pp. 1–10.
- [18] S. Huang, Z. Gojcic, Z. Wang, F. Williams, Y. Kasten, S. Fidler, K. Schindler, and O. Litany, “Neural lidar fields for novel view synthesis,” in *IEEE/CVF ICCV*, 2023, pp. 18 236–18 246.
- [19] A. Ramazzina, S. Walz, P. Dahal, M. Bjelic, and F. Heide, “Gated fields: Learning scene reconstruction from gated videos,” in *Proceedings of the IEEE/CVF CVPR*, 2024, pp. 10 530–10 541.
- [20] A. Ramazzina, M. Bjelic, S. Walz, D. Scheuble, and F. Heide, “Scatternerf: Seeing through fog with physically-based inverse neural rendering,” in *IEEE/CVF ICCV*, 2023, pp. 17 957–17 968.
- [21] Y. Qiao, M. Shao, L. Meng, and K. Xu, “Restorgs: Depth-aware gaussian splatting for efficient 3d scene restoration,” in *CVPR*, 2025.
- [22] X. Guo, Y. Li, and H. Ling, “Lime: Low-light image enhancement via illumination map estimation,” *IEEE Transactions on image processing*, vol. 26, no. 2, pp. 982–993, 2016.
- [23] Y. Jiang, X. Gong, D. Liu, Y. Cheng, C. Fang, X. Shen, J. Yang, P. Zhou, and Z. Wang, “Enlightengan: Deep light enhancement without paired supervision,” *IEEE transactions on image processing*, vol. 30, 2021.
- [24] C. Wei, W. Wang, W. Yang, and J. Liu, “Deep retinex decomposition for low-light enhancement,” *arXiv preprint arXiv:1808.04560*, 2018.
- [25] C. Guo, C. Li, J. Guo, C. C. Loy, J. Hou, S. Kwong, and R. Cong, “Zero-reference deep curve estimation for low-light image enhancement,” in *Proceedings of the IEEE/CVF CVPR*, 2020, pp. 1780–1789.
- [26] D. Feijoo, J. C. Benito, A. Garcia, and M. V. Conde, “Darkir: Robust low-light image restoration,” in *CVPR*, 2025, pp. 10 879–10 889.
- [27] Y. Dai, C. Li, S. Zhou, R. Feng, Y. Luo, and C. C. Loy, “Flare7k++: Mixing synthetic and real datasets for nighttime flare removal and beyond,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 11, pp. 7041–7055, 2024.
- [28] Y. Zhou, D. Liang, S. Chen, S.-J. Huang, S. Yang, and C. Li, “Improving lens flare removal with general-purpose pipeline and multiple light sources recovery,” in *Proceedings of the IEEE/CVF ICCV*, 2023.
- [29] S. Ye, Z.-H. Dong, Y. Hu, Y.-H. Wen, and Y.-J. Liu, “Gaussian in the dark: Real-time view synthesis from inconsistent dark images using gaussian splatting,” in *Computer Graphics Forum*, vol. 43, no. 7, 2024.
- [30] B. Mildenhall, P. Hedman, R. Martin-Brualla, P. P. Srinivasan, and J. T. Barron, “Nerf in the dark: High dynamic range view synthesis from noisy raw images,” in *IEEE/CVF CVPR*, 2022, pp. 16 190–16 199.
- [31] K. Zhang, G. Riegler, N. Snavely, and V. Koltun, “Nerf++: Analyzing and improving neural radiance fields,” *preprint arXiv:2010.07492*, 2020.
- [32] J. T. Barron, B. Mildenhall, D. Verbin, P. P. Srinivasan, and P. Hedman, “Mip-nerf 360: Unbounded anti-aliased neural radiance fields,” in *Proceedings of the IEEE/CVF CVPR*, 2022, pp. 5470–5479.
- [33] Z. Yang, Y. Chen, J. Wang, S. Manivasagam, W.-C. Ma, A. J. Yang, and R. Urtasun, “Unisim: A neural closed-loop sensor simulator,” in *Proceedings of the IEEE/CVF CVPR*, 2023, pp. 1389–1399.
- [34] T. Fischer, L. Porzi, S. Rota Bulò, M. Pollefeys, and P. Kotschieder, “Multi-level neural scene graphs for dynamic urban environments,” in *Proceedings of the IEEE/CVF CVPR*, 2024.
- [35] S. Singh, A. Garg, and K. Mitra, “Hdrsplat: Gaussian splatting for high dynamic range 3d scene reconstruction from raw images,” *arXiv preprint arXiv:2407.16503*, 2024.
- [36] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. J. Black, “Smpl: A skinned multi-person linear model,” in *Seminal Graphics Papers: Pushing the Boundaries, Volume 2*, 2023, pp. 851–866.
- [37] Y. Jiang, J. Tu, Y. Liu, X. Gao, X. Long, W. Wang, and Y. Ma, “Gaussianshader: 3d gaussian splatting with shading functions for reflective surfaces,” in *IEEE/CVF CVPR*, 2024, pp. 5322–5332.
- [38] Z. Bi, Y. Zeng, C. Zeng, F. Pei, X. Feng, K. Zhou, and H. Wu, “Gs3: Efficient relighting with triple gaussian splatting,” in *SIGGRAPH Asia 2024 Conference Papers*, 2024, pp. 1–12.
- [39] F. Heide, “Structure-aware computational imaging,” Ph.D. dissertation, University of British Columbia, 2016.
- [40] C. Poynton, *Digital video and HD: Algorithms and Interfaces*. Elsevier, 2012.
- [41] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, “Image quality assessment: from error visibility to structural similarity,” *IEEE transactions on image processing*, vol. 13, no. 4, pp. 600–612, 2004.
- [42] Y. Bin, W. Hu, H. Wang, X. Chen, and B. Wang, “Normalcrafter: Learning temporally consistent normals from video diffusion priors,” *arXiv preprint arXiv:2504.11427*, 2025.
- [43] M. Turkulainen, X. Ren, I. Melekhov, O. Seiskari, E. Rahtu, and J. Kannala, “Dn-splatter: Depth and normal priors for gaussian splatting and meshing,” in *2025 IEEE/CVF WACV*, pp. 2421–2431.
- [44] H. e. a. Caesar, “nuscenes: A multimodal dataset for autonomous driving,” in *IEEE/CVF CVPR*, 2020, pp. 11 621–11 631.
- [45] P. Sun, H. Kretschmar, X. Dotiwalla, A. Chouard, V. Patnaik, P. Tsui, J. Guo, Y. Zhou, Y. Chai, B. Caine *et al.*, “Scalability in perception for autonomous driving: Waymo open dataset,” in *Proceedings of the IEEE/CVF CVPR*, 2020, pp. 2446–2454.
- [46] J. Chung, J. Oh, and K. M. Lee, “Depth-regularized optimization for 3d gaussian splatting in few-shot images,” in *IEEE/CVF CVPR*, 2024.
- [47] Q. Wang, L. Fan, Y. Wang, Y. Chen, and Z. Zhang, “Freevs: Generative view synthesis on free driving trajectory,” *arXiv preprint arXiv:2410.18079*, 2024.